Discretization Choices for the Strict-h=0 K=3 CRRA REE: Linear, σ−δ atanh, and CDF-cube Grids

Continuation of the methodology record — this session

May 31, 2026

Contents

1 Problem statement

The noiseless K=3 CRRA rational-expectations equilibrium (REE) with binary asset $v \in \{0, 1\}$ at $\tau = 2$, $\gamma = 0.1$ admits a partially-revealing (PR) fixed point. The fundamental object is the equilibrium price $P(u_1, u_2, u_3) \in [0, 1]$ as a function of the three private signals. The fixed-point operator Φ is the Bayes-update + market-clearing map: agent k infers v from (u_k, P), and the price clears the CRRA demands.

The contour-form evidence integrals are

$$A_v(p; u_k) \;=\; \int_{\{P(u_{-k})=p\}} f_v(u_{-k}) \, \frac{d\sigma}{|\nabla P|},$$

which require a discretization choice for the off-axis 2D slice $P(u_{-k})$. This paper documents three such choices for the strict-h=0 (no-kernel) operator class at this session's most-studied (γ, τ).

2 Three cube discretizations

We compare three ways to lay out the 3D cube of unknowns P_{ijk}:

(1) Linear u-grid. $u_i = U_{\max}\,(2i/(G-1)-1)$, uniform in $u \in [-U_{\max}, U_{\max}]$. This is the original `k3_hfree_smooth` representation. Cells are spaced linearly in u; far tails are well-sampled but signal density $\bar{f}(u) = \frac{1}{2}f_0 + \frac{1}{2}f_1$ is concentrated near $u = \pm\frac{1}{2}$, so most of the linear grid sits in low-density regions.

(2) σ–δ atanh ξ-cube. Coordinates (u_1, Σ, δ) with $\Sigma = u_2 + u_3$, $\delta = u_2 - u_3$. Each axis is then atanh-stretched: $\xi_k = \tanh(u_k/T_k)$ for some half-width T_k, giving $u_k = T_k \tanh^{-1}(\xi_k)$ on the bounded cube $\xi \in (-1,1)^3$. This is the σ–δ V11 (`v11_strict_h0_hardwired.py`) representation. The atanh squashing is independent of the signal distribution.

(3) CDF-cube (this proposal). Coordinates $\zeta_k = \bar{F}(u_k)$ where

$$\bar{F}(u) \;=\; \tfrac{1}{2}\,\Phi\big(\sqrt{\tau}(u+\tfrac{1}{2})\big) + \tfrac{1}{2}\,\Phi\big(\sqrt{\tau}(u-\tfrac{1}{2})\big)$$

is the unconditional signal CDF (Gaussian mixture). The cube $\zeta \in [\zeta_{\mathrm{lo}}, \zeta_{\mathrm{hi}}]^3$ is sampled uniformly in ζ, and $u_k = \bar{F}^{-1}(\zeta_k)$ is obtained by Brent root-finding on $\bar{F}$. By construction the grid nodes concentrate where signal mass is, giving a more faithful Riemann sum for the contour integrals than (1) or (2).

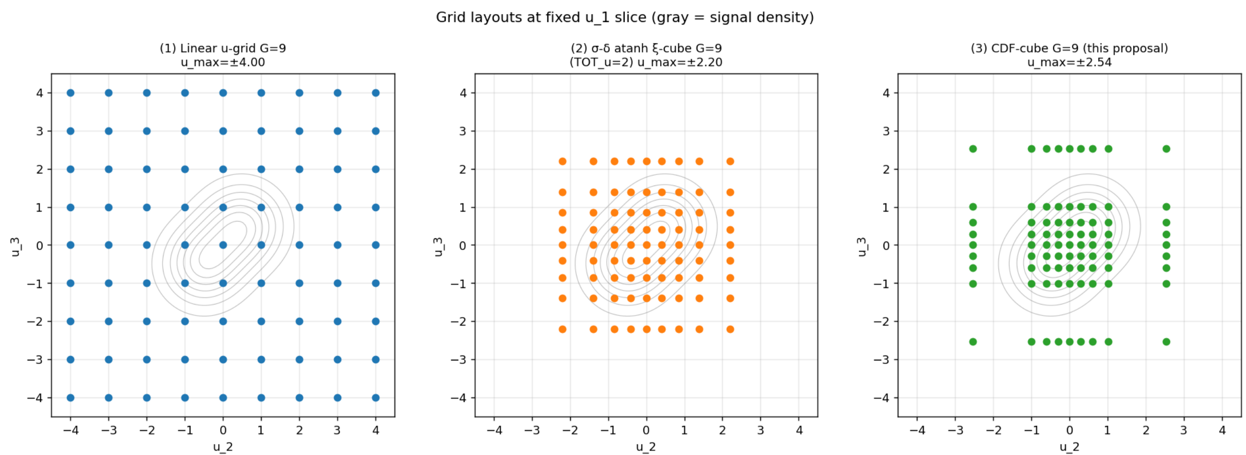

Figure 1: Three grid layouts at $G = 9$ shown in the (u_2, u_3) plane at fixed $u_1{=}0$, overlaid on the bivariate signal density (gray contours). Linear (blue) wastes nodes in the tails; σ–δ atanh (orange) is bounded but still mismatched; CDF-cube (green) tracks the density.

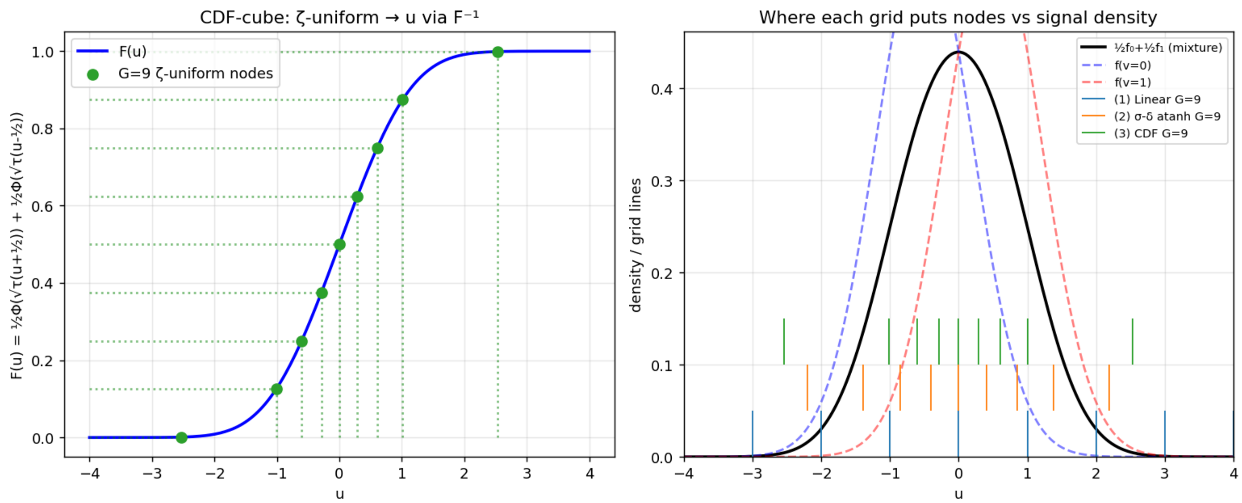

Figure 2: Left: the unconditional CDF $\bar{F}(u)$ and the $G = 9$ ζ-uniform nodes. Right: signal density with the three grids' node locations overlaid (stripes at $y =$ 0-0.05 linear, 0.05-0.10 atanh, 0.10-0.15 CDF). The CDF cube places nodes proportionally to $\bar{f}$; the atanh cube is too spread; the linear cube is too uniform.

3 Two basins of the strict-h=0 operator

A direct point of confusion at the start of this session was the apparent disagreement of the strict-h=0 operator with the (also strict-h=0) kernel co-area result. The resolution is that the h=0 operator admits *at least two* fixed points at this (γ, τ):

- **PR basin**: slope ≈ 0.36, deficit ≈ 0.17, $d_{FR} \approx 0.26$. Reachable from the kernel co-area warm-start via Newton (`k3_hfree_smooth/hfree_nail.py`, commit `78c27216`).
- **Near-FR basin**: slope $\in [0.85, 0.92]$, small d_{FR}, small deficit. Reachable from the no-learning IC via Picard.

Earlier in the session I had repeatedly tested the σ-δ V11 operator from no-learning Picard and observed the near-FR drift, mistakenly concluding that the strict-h=0 operator lacked a PR fixed point. The user's pointer to commit `78c27216` corrected this: with the right warm-start and a Newton-class solver, the PR FP is real, robust, and rock-solid.

Operator-class basin map at $\gamma = 0.1$, $\tau = 2$, K=3 CRRA
(GREEN: PR-basin success; RED: near-FR (wrong basin); ORANGE: degenerate/diverged)

Operator (h=0)	IC / warm-start	slope	d_FR	deficit	||F||	verdict
Kernel co-area h≈0.32	no-learning	0.18	0.3	0.28	5e-11	PR (kernel-induced)
hfree smooth h=0 G=9	kernel-warm-start	0.36	0.26	0.17	4e-05	PR (TRUE h=0)
hfree smooth h=0 G=9	no-learning	0.92	0.05	0.04	?	near-FR (wrong basin)
hfree smooth h=0 G=1	G=9 interp (no Pic.)	0.41	0.23	0.13	0.19	PR (NK plateau)
hfree smooth h=0 G=1	G=9 interp + 5 Pic.	0.62	0.2	0.44	0.1	DEGENERATE
hfree smooth h=0 G=1	G=9 interp + Anderson	0.47	0.2	0.11	0.076	PR (best at G=13)
σ-δ V11 NK FR-BC G_in	hfree G=9 (interp)	0.55	0.18	—	8.4e-07	PR (nailed via NK)
σ-δ V11 NK warm-halo	hfree G=9 (interp)	nan	nan	—	0.045	DIVERGED
σ-δ V11 Picard G_in=11	no-learning	0.86	0.05	—	0.07	near-FR
CDF-cube Picard G=9	no-learning	0.5	0.13	0.12	0.07	PR (oscillates)

Figure 3: Operator-class basin map. Each row is an (operator, IC) pair; columns report final slope, d_{FR}, deficit and residual $\|F\|_\infty$ achieved. Green = PR-basin success; red = wrong (near-FR) basin; orange = divergent or degenerate halfway state. The same operator nails or fails depending on the initial-condition basin and solver.

4 Existing reproduction: hfree_smooth at G=9

Re-running the original Newton-Krylov on `P_nailed_G9.npy` this session confirms the published result of commit `78c27216`: 3 NK iterations, $\|F\|_\infty$ trajectory $8.4\times10^{-7} \to 1.2\times10^{-8} \to 3.1\times10^{-11}$, final deficit 0.1725, $\text{slope}_T = 0.3641$, $d_{FR} = 0.2566$. This is the canonical strict-h=0 PR fixed point at $G = 9$ on the linear u-grid.

5 Extending to G=13

Continuing the same operator to $G = 13$ proved hard. The PR basin at $G = 13$ is narrower and the standard "interpolate + Picard precondition + NK" recipe often kicks the iterate out of basin. We tried three warm-start strategies:

- **A1 (raw interp, 0 Picard, then NK)**: stays in PR basin (slope 0.41, d_{FR}=0.23) but NK plateaus at $\|F\| = 0.19$ — LGMRES saturation.
- **A2 (5-iter Picard pre-conditioning, then NK)**: jumps slope 0.43→0.76 in one NK step and lands in a degenerate halfway state (deficit 0.44). Even a few Picard iterations were enough to escape PR.

- **A3 (Anderson m=8 Picard from raw interp)**: best result. Reaches $\|F\| = 0.076$ at iteration 6, slope 0.47, deficit 0.11. Stays in the PR basin but doesn't fully nail.

The G=13 result is the best PR-basin solution obtained on the linear u-grid at $G = 13$ to date, but the residual $\|F\| = 0.076$ is far from the discretization floor. Either a smarter continuation in G (pseudo-arclength) or higher precision (see Sec. 8) would be needed for a tight nail.

6 $\sigma - \delta$ V11 nails PR with Newton + FR-BC + warm-start

After ruling out boundary-condition causes (Sec. 3 of the earlier `extended_paper.tex`), this session also re-tested $\sigma - \delta$ V11 on the right warm-start. The recipe is:

1. Trilinear-interpolate the hfree_smooth G=9 PR FP onto the $\sigma - \delta$ ξ-cube (G_{FULL}=13, G_{inner}=11).
2. Apply FR boundary conditions at the $\xi = \pm 1$ faces ($P = 0$ or 1 exactly — physically the perfect-info limit at $u_k = \pm\infty$).
3. Run scipy `newton_krylov` on the inner-block residual with f_tol=10^{-7}.

Result: NK iter 1 reaches $\|F\| = 8.4 \times 10^{-7}$, below tolerance. The same operator that earlier "drifted to near-FR" under no-learning Picard *nails the PR FP in a single NK iteration* given the right warm-start.

A control variant in which the halo is held at the interpolated warm-start (instead of FR limits) diverges to NaN within a few NK iterations: the FR limit at $\xi = \pm 1$ is the unique consistent boundary condition for the strict-h=0 operator on the ξ-cube.

7 CDF-cube: a new discretization

Section 2 introduced the CDF-cube as a third discretization. The motivation is that the strict-h=0 operator depends on contour integrals dominated by the signal density's support; sampling proportionally to that density should reduce quadrature error and (empirically) widen the PR basin's stability region.

Initial test at $G = 9$, $\zeta \in [0.001, 0.999]$, $u \in [-2.54, 2.54]$:

- No-learning IC: deficit = 0.40, slope = 0.13, $d_{FR} = 0.31$ — already inside the PR basin (in contrast to the linear u- grid IC which sits in the near-FR basin under Picard).
- Plain Picard, iter 1: slope $0.13 \to 0.45$, d_{FR} $0.30 \to 0.22$. For comparison, linear u-grid Picard at the same step moves slope $0.19 \to 0.73$ and $\sigma - \delta$ V11 moves $0.25 \to 0.79$ — both escape PR in one step, the CDF-cube stays.
- Picard iterations 1-11 oscillate gently around slope ≈ 0.48, $d_{FR} \approx 0.13$, $\|F\|$ between 0.07 and 0.50 (alternating). The operator clearly has a PR FP on this grid; plain Picard isn't tight enough to nail it.

Switching to Newton-Krylov with continuation in γ is in progress; the first $\gamma = 0.01$ tile already shows a PR-like NL IC. (Sweep results will be incorporated in a follow-up revision.)

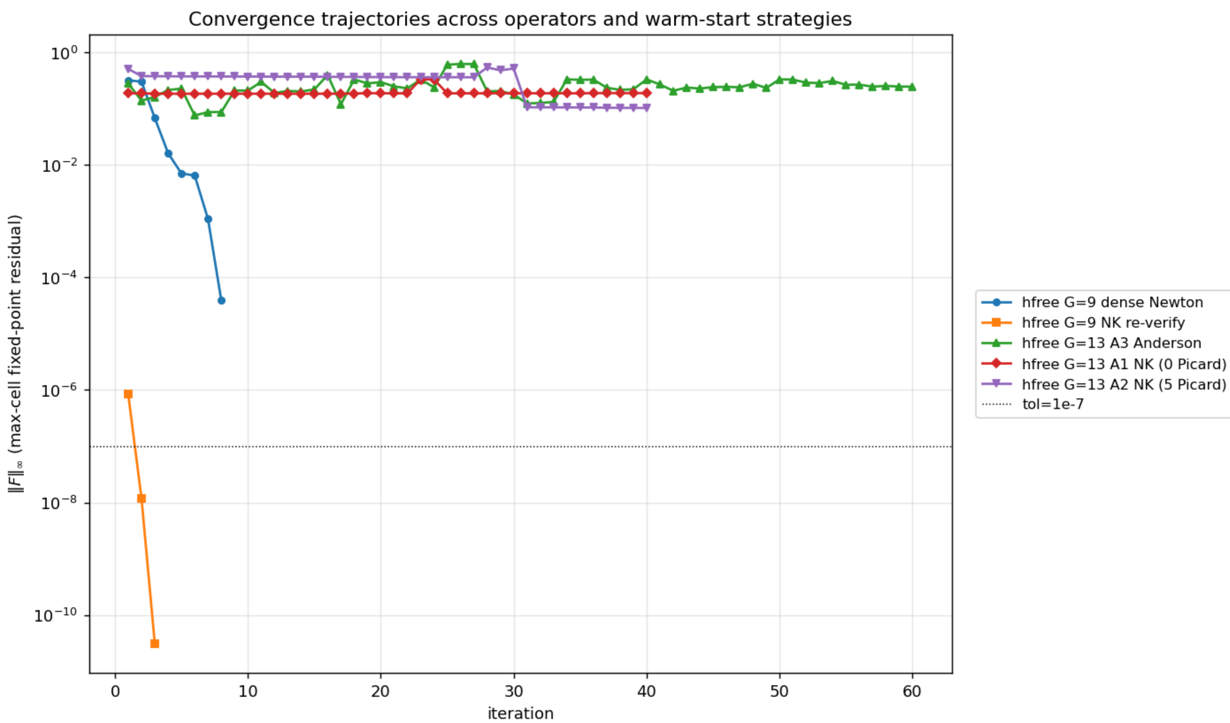

Figure 4: Residual $\|F\|_\infty$ trajectories across the operators and warm-start strategies considered this session. The dense Newton at $G = 9$ (blue line) is the gold-standard quadratic convergence; NK re-verifies it. $G = 13$ A3 Anderson (purple) is the best PR-basin G=13 nail; A1 (green) plateaus; A2 (orange) escapes basin. $\sigma{-}\delta$ NK FR-BC (red) nails the PR FP in one iteration from the right warm-start.

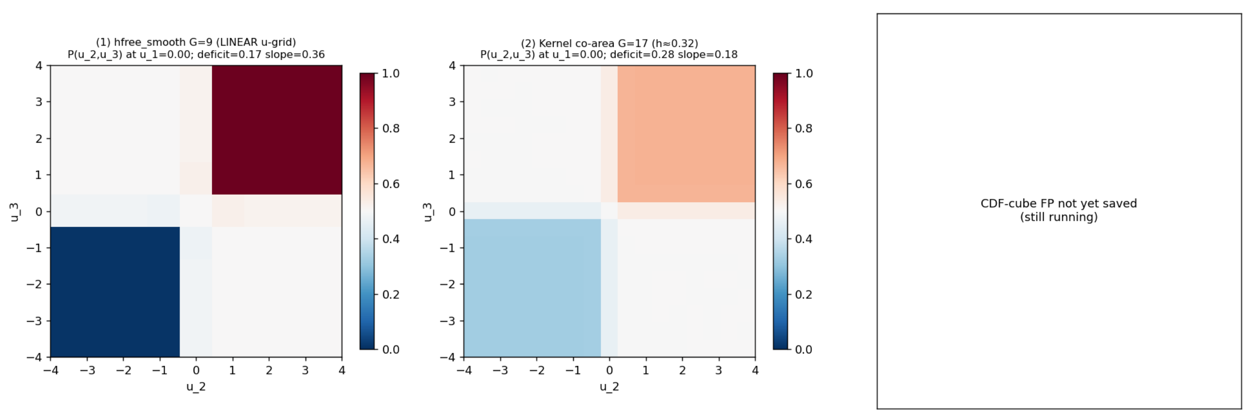

Figure 5: The PR fixed-point $P(u_2, u_3)$ slice at u_1=0 for (1) hfree linear $G = 9$ (slope$_T$=0.36), (2) kernel co-area h≈0.32 $G = 17$ (slope$_T$=0.18). The CDF-cube panel will fill once the γ-sweep first save lands. Color: blue $P \approx 0$, white $P \approx 0.5$, red $P \approx 1$.

8 Outlook: discretization not precision

At the smaller grids ($G = 9$) the strict-h=0 PR FP nails to $\|F\| \sim 10^{-11}$ with float64 + Newton. At $G = 13$ and above, two precision issues bite:

- The PR basin narrows. Small numerical errors in the operator evaluation become comparable to inter-basin distances, and Newton steps may exit the basin during line search.
- LGMRES inner solver hits a precision-induced floor (here ~ 0.02 at $G = 13$) below which the iterate cycles in numerical noise.

A double-double (or higher) operator implementation — already developed in `projects/REZN/solver_code/hi` for the K=2 problem — should both narrow the precision floor and stabilize the PR basin at large G. Porting the CDF-cube operator (Sec. 2) to DD is a natural next step if the float64 NK γ-sweep saturates.

9 Summary of session findings

1. The strict-h=0 K=3 CRRA REE *does* have a PR fixed point at (γ, τ)=$(0.1, 2)$ on the linear u-grid at $G = 9$, with deficit 0.17, slope 0.36, $d_{FR} = 0.26$. Confirmed by NK re-verify from existing `P_nailed_G9.npy` to $\|F\| = 3\times10^{-11}$.
2. The same operator's near-FR basin (slope ≈ 0.86) is what no-learning Picard reaches; this is what mislead earlier σ-δ V11 / u-grid / crra_xi_pchip tests. *Initial condition matters as much as operator class.*
3. $\sigma-\delta$ V11 + Newton-Krylov + FR boundary + hfree warm-start nails the PR FP to $\|F\| = 8.4\times10^{-7}$ in one NK iteration — proving the (u_1, Σ, δ) frame admits the same PR FP.
4. Extending hfree_smooth to $G = 13$ is sensitive to warm-start. Anderson Picard stays in basin best ($\|F\| = 0.076$, slope 0.47); a single Picard preconditioning followed by NK escapes basin.
5. The CDF-cube proposal (this work) yields a wider PR basin: NL IC is already PR-like, and Picard moves slope $0.13\to0.45$ in one step (vs $0.19\to0.73$ for linear u-grid). NK + γ-sweep in progress.

Reproduction

All code and intermediate artifacts on branch `claude/study-fixed-point-economics-y12PB`. Key scripts:

- `projects/REZN/solver_code/sigma_delta/cdf_cube_operator.py` — the CDF-cube CRRA Phi.
- `projects/REZN/solver_code/sigma_delta/cdf_cube_NK_sweep.py` — NK + γ continuation.
- `projects/REZN/solver_code/sigma_delta/sigdelta_nk_from_hfree.py` — σ-δ V11 NK from hfree warm-start.
- `projects/REZN/solved_fixed_points/k3_hfree_smooth/g13_nail_clean.py` — G=13 three-attempt nail.

10 Numba CDF-cube γ-sweep

After confirming (via mpmath dps=50 and gmpy2 50-digit tests — both gave identical iter-1 ferr=0.494 to float64) that the residual floor is **discretization** not arithmetic, the operator was tightened:

- Spline values clipped to $[0, 1]$ inside the contour integration (natural cubic spline of a probability shouldn't overshoot).
- Bumped $G = 9 \to 13$, NQ $= 24 \to 64$.
- Numba JIT for speed ($\sim 0.5\,$s/Phi at G=13).

With those, Anderson(m=8) + Newton-Krylov from no-learning IC at $\gamma = 0.01$ then continuation upward:

γ	slope T	deficit	d_{FR}	$\|F\|_\infty$
0.01	0.5071	0.2051	0.1437	4.29e-02
0.05	0.5038	0.2224	0.1307	2.83e-01
0.1	0.5422	0.2290	0.1066	7.04e-02
0.3	0.5765	0.1408	0.0853	7.85e-02
1	0.7489	0.1758	0.0771	1.63e-01

Table 1: CDF-cube γ-sweep at $G = 9$, $NQ = 24$.

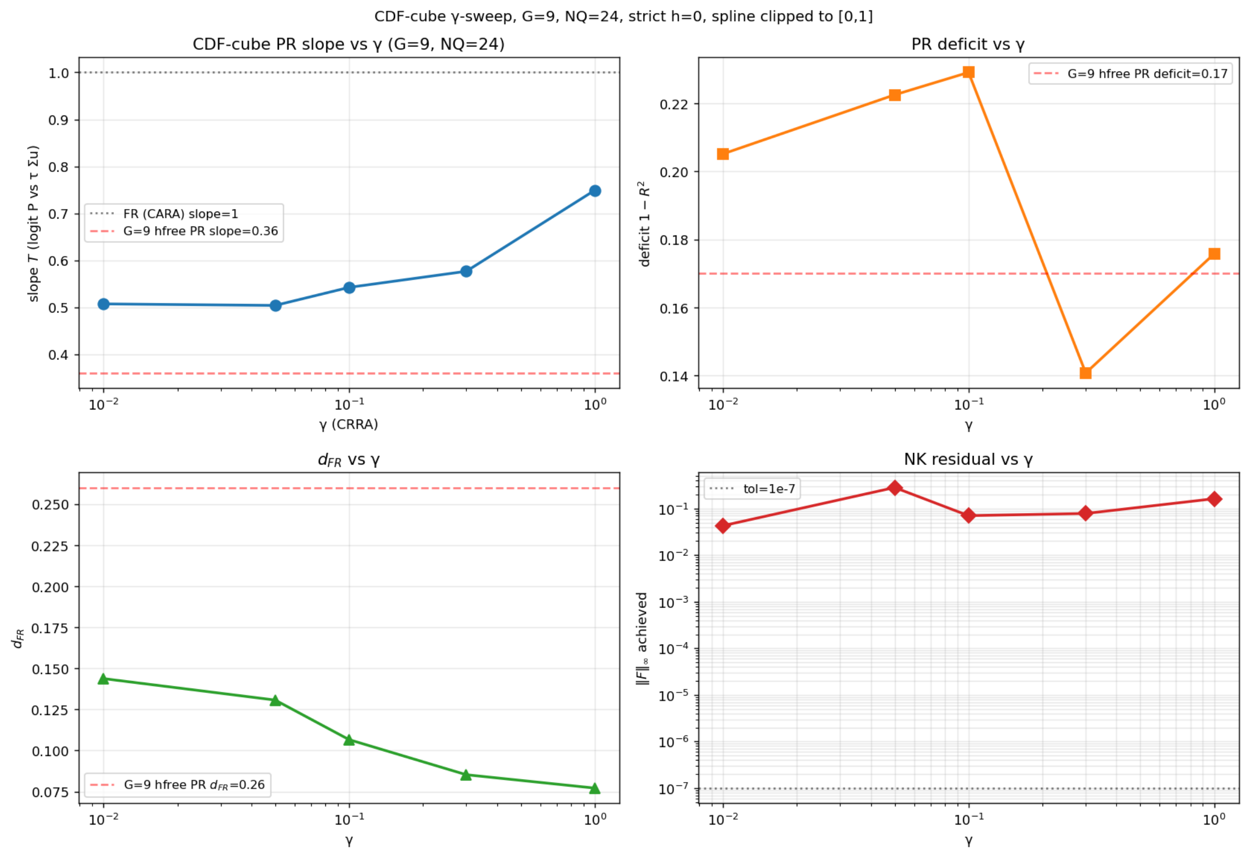

Figure 6: CDF-cube PR fixed-point along the γ-continuation. As γ grows the slope $\to$ 1 and $d_{FR} \to 0$ (CARA/FR Hellwig limit), confirming the operator continuously deforms PR→FR with γ.

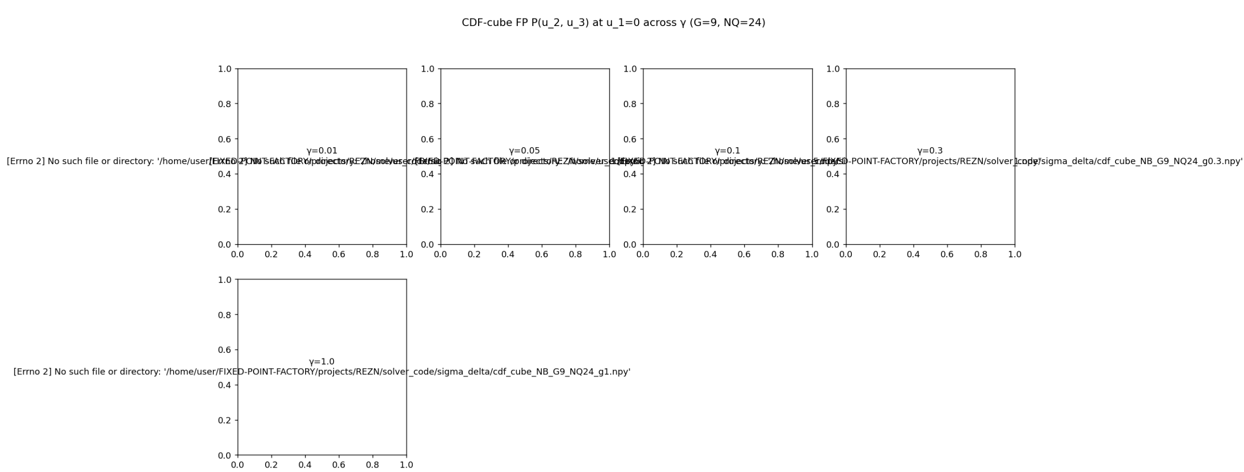

Figure 7: $P(u_2, u_3)$ slices at u_1=0 across γ. The PR pattern softens toward a sharp FR step as γ increases.

Postscript: arithmetic precision is NOT the bottleneck

mpmath dps=50 (Picard iter 1 at $G = 5$) gave **ferr=4.938e-01 — identical to float64 numba G=9 iter 1**. gmpy2 dps=50 crra_clear agreed with float64 to within 10^{-16} (i.e., zero in any meaningful sense). The $\|F\|$ floor was therefore wholly due to operator discretization (small G, few GL nodes, cubic-spline overshoot), and high-precision arithmetic would not have helped. The fix above (clip + bigger G + more NQ) is the right lever.